Build Web Apps You Can Formally Verify.

A Rust web framework that proves security-critical logic with Lean 4.

AI Writes Code Faster Than Anyone Can Verify It.

Tests ∃

Find some bugs.

Proofs ∀

Rule out a whole class.

> One missed edge case leaks another customer's data.

Security-Critical Backend Teams.

- Multi-tenant SaaS
- Fintech and payments
- AI-written backends

> Where one authorization bug is a breach.

04 / SOLUTION

Write Rust. Prove It in Lean 4.

01 · RUST

Application logic

→

02 · LEAN 4

Machine-checked proof

→

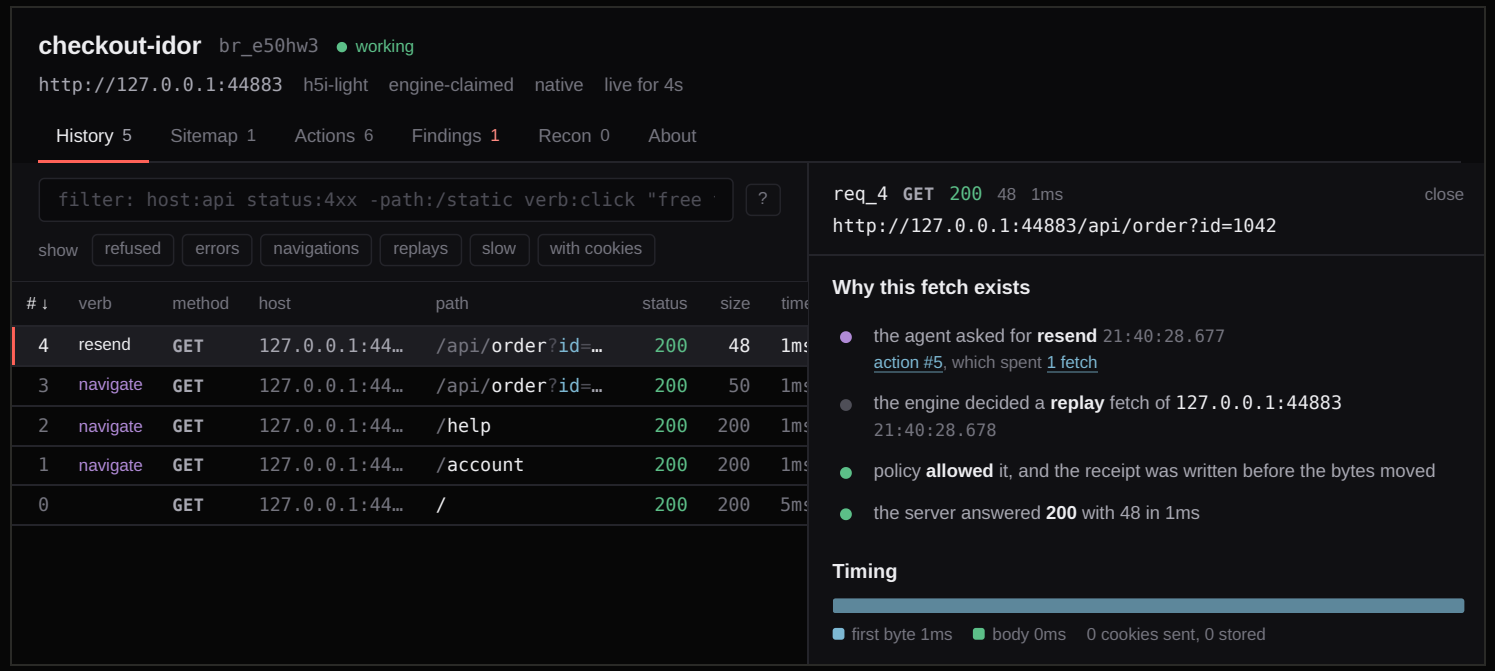
03 · GUARANTEE

Authorization always holds

Proofs cover stated properties under explicit assumptions

Prove the Code. Test the System.

Red-teaming covers what proofs can't. Every finding becomes a CI test.



Actual h5i ui, local demo

Early Open-Source Traction.

900+

GITHUB STARS

3.8k+

DOWNLOADS

Start With Rust. Expand to Every Web App.

WEDGE

Rust backend teams

EXPANSION

Any web app, via red-teaming

LONG TERM

Verifying all AI-written code

Open-Source Core. Paid Verification.

FREE

Framework, proofs, CLI

PAID · PLANNED

CI guarantee reports, audit trails, support

Hideaki Takahashi

Columbia CS PhD. Software security and formal methods.

IEEE S&P 2026

ACM CCS 2026

Creator of h5i

Building Verifiable Software.

Raising a seed round to bring formal verification into production.

Start a conversation ↗

Explore h5i ↗